

GSB518 Notes

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Jul 10, 2026

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Preface

This is a selection of Handouts for Cal Poly GSB 518, Essential Statistics for Business Analytics, covering topics in Probability and Statistics.

1 Randomness and Probability

Probability comes up in a wide variety of situations. Consider just a few examples.

1. The probability that you roll doubles in a turn of a board game is 0.167.
2. The probability that a randomly selected Cal Poly student is a California resident is 0.84
3. The probability that the high temperature in San Luis Obispo, CA tomorrow is above 90 degrees F is 0.4
4. The probability that the Los Angeles Dodgers win the next World Series is 0.27.
5. The probability that extraterrestrial life currently exists somewhere in the universe.
6. The probability that you ate an apple on April 17, 2019.

Example 1.1. How are the situations above similar, and how are they different? What is one feature that all of the situations have in common? Is the interpretation of “probability” the same in all situations? The goal here is to just think about these questions, and not to compute any probabilities (or to even think about how you would).

- A phenomenon is **random** if there are multiple potential outcomes, and there is **uncertainty** about which outcome will occur.
- Uncertainty is understood in broad terms, and in particular does not only concern future occurrences.
- Many phenomena involve physical randomness, like flipping a coin or drawing powerballs at random from a bin, or in statistical applications of random sampling or random assignment.
- But in many other situations, randomness just vaguely reflects uncertainty.
- Random does *not* mean haphazard. In a random phenomenon, while individual outcomes are uncertain, there is a *regular distribution of outcomes over a large number of (hypothetical) repetitions*.
- Also, random does *not* necessarily mean equally likely. In a random phenomenon, certain outcomes or events might be more or less likely than others.

- The **probability** of an event associated with a random phenomenon is a number in the interval $[0, 1]$ measuring the event's likelihood or degree of uncertainty. A probability can take any value in the continuous scale from 0% to 100%.
- There are two main interpretations of probability.
 - **Long run relative frequency.** The probability of an event can be interpreted as the proportion of times that the event would occur in a very large number of hypothetical repetitions of the random phenomenon.
 - **Subjective probability.** There are many situations where the outcome is uncertain, but it does not make sense to consider the situation as repeatable. In such situations, a subjective (a.k.a., personal) probability describes the degree of likelihood a given individual ascribes to a certain event. Think of subjective probabilities as measuring relative degrees of likelihood rather than long run relative frequencies.
- Fortunately, the mathematics of probability work the same way regardless of the interpretation. In either case, the same basic logical consistency requirements must be satisfied.
- A **simulation** involves an artificial recreation of the random phenomenon, usually using a computer. The probability of an event can be approximated by simulating the random phenomenon a large number of times and determining the proportion of simulated repetitions on which the event occurred out of the total number of repetitions in the simulation.

Example 1.2. One of the oldest documented problems in probability is the following: If three fair six-sided dice are rolled, what is more likely: a sum of 9 or a sum of 10?

1. Conduct a *simulation* to investigate this question.
2. Use the simulation results to approximate the probability that the sum is 9; repeat for a sum of 10.
3. It can be shown that the theoretical probability that the sum is 9 is $25/216 = 0.116$. Write a clearly worded sentence interpreting this probability as a long run relative frequency.

4. It can be shown that the theoretical probability that the sum is 10 is $27/216 = 0.125$. How many times more likely is a sum of 10 than a sum of 9?

- A probability can be interpreted as a theoretical long run relative frequency.
- A probability can be approximated by a relative frequency from a large number of simulated repetitions, but there is some simulation margin of error, due to natural variability in the simulation.
- The margin of error when approximating a single probability based on a simulated relative frequency is roughly on the order $1/\sqrt{N}$, where N is the *number of independently simulated values used to calculate the relative frequency*. For example, if $N = 10000$ then the margin of error is roughly $1/\sqrt{10000} = 0.01$. (But be careful when approximating *conditional* probabilities.)

Example 1.3. What is your subjective probability that Professor Ross has ever attended a “famous” sporting event (like the Superbowl, a World Series game 7, a World Cup match, an NCAA tournament game with a famous shot, an Olympic final, etc)? Consider the following two bets, and suppose you must choose only one.

- A. You win \$100 if Professor Ross has attended a famous sporting event, and you win nothing otherwise.
 - B. A box contains 40 green and 60 gold marbles that are otherwise identical. The marbles are thoroughly mixed and one marble is selected at random. You win \$100 if the selected marble is green, and you win nothing otherwise.
1. Which of the above bets would you prefer? Or are you completely indifferent? What does this say about your subjective probability that Professor Ross has attended a famous sporting event?
2. If you preferred bet B to bet A, consider bet C which has a similar setup to B but now there are 20 green and 80 gold marbles. Do you prefer bet A or bet C? What does this say about your subjective probability that Professor Ross has attended a famous sporting event?

3. If you preferred bet A to bet B, consider bet D which has a similar setup to B but now there are 60 green and 40 gold marbles. Do you prefer bet A or bet D? What does this say about your subjective probability that Professor Ross has attended a famous sporting event?
4. Continue to consider different numbers of green and gold marbles. Can you zero in on your subjective probability?

Example 1.4. Suppose your subjective probabilities for who will be the next World Series champion satisfy the following conditions.

- The Phillies and Mets are equally likely to win
 - The Yankees are 1.5 times more likely than the Phillies to win
 - The Dodgers are 2 times more likely than the Yankees to win
 - The winner is as likely to be among these four teams—Phillies, Mets, Yankees, Dodgers—as not
1. Compute the subjective probability that each team wins.
 2. Represent these probabilities in a spinner, like from a kids game.
 3. Compute the probability that the winner is not the Dodgers.

4. How many times more likely are the Dodgers to not win than to win? (This is referred to as the “odds”.)

5. Compute the probability that the winner is the Dodgers or the Yankees.

- We will use the two interpretations — long run relative frequencies and subjective probabilities — interchangeably.
- With subjective probabilities it is often helpful to consider what might happen in a simulation.
- It is also useful to consider long run relative frequencies in terms of relative degrees of likelihood.
- Fortunately, the mathematics of probability work the same way regardless of the interpretation.
- A probability takes a value in the sliding scale from 0 to 100%.
- Don’t just focus on computation; always remember to properly interpret probabilities.

Example 1.5. In each of the following parts, which of the two probabilities, a or b, is larger, or are they equal? You should answer conceptually without attempting any calculations. Explain your reasoning.

1. Flip a coin *which is known to be fair* 10 times.

- a. The probability that the results are, in order, HHHHHHHHHH.
- b. The probability that the results are, in order, HHTHTTTTHHT.

2. Flip a coin which is known to be fair 10 times.

- a. The probability that all 10 flips land on H.
- b. The probability that exactly 5 flips land on H.

- Warning! Your psychological judgment of probabilities is often inconsistent with the mathematical logic of probabilities.
- When interpreting probabilities, consider the conditions under which the probabilities were computed, in the proper direction

- When interpreting probabilities, be careful not to confuse “the particular” with “the general”.
 - “**The particular:**” A very specific event, surprising or not, often has low probability.
 - “**The general:**” While a very specific event often has low probability, if there are many like events their combined probability can be high.
- Even if an event has extremely small probability, given enough repetitions of the random phenomenon, the probability that the event occurs on *at least one* of the repetitions is often high.

1.1 Exercises

Exercise 1.1. In each of the following parts, which of the two probabilities, a or b, is larger, or are they equal? You should answer conceptually without attempting any calculations. Explain your reasoning.

1. Consider a Cal Poly student who frequently has blurry, bloodshot eyes, generally exhibits slow reaction time, always seems to have the munchies, and disappears at 4:20 each day. Which of the following events, *A* or *B*, has a *higher* probability? (Assume the two probabilities are not equal.)
 - a. The student has a GPA above 3.0.
 - b. The student has a GPA above 3.0 and smokes marijuana regularly.
2. Randomly select a man.
 - a. The probability that a randomly selected man is greater than six feet tall.
 - b. The probability that a randomly selected man who plays in the NBA is greater than six feet tall.
3. In the [Powerball lottery](#) there are roughly 292 million possible winning number combinations, all equally likely.
 - a. The probability you win the next Powerball lottery if you purchase a single ticket, 4-8-15-16-42, plus the Powerball number, 23
 - b. The probability you win the next Powerball lottery if you purchase a single ticket, 1-2-3-4-5, plus the Powerball number, 6.
4. Continuing with the Powerball
 - a. The probability that the numbers in the winning number are not in sequence (e.g., 4-8-15-16-42-23)
 - b. The probability that the numbers in the winning number are in sequence (e.g., 1-2-3-4-5-6)

5. Continuing with the Powerball

- a. The probability that you win the next Powerball lottery if you purchase a single ticket.
- b. The probability that someone wins the next Powerball lottery. (FYI: especially when the jackpot is large, there are hundreds of millions of tickets sold.)

Exercise 1.2. Your favorite local weatherperson forecasts a 30% chance of rain tomorrow and a 60% chance of rain the next day in your city.

1. Explain how these probabilities are subjective.
2. You ask Donny Don't to interpret the 30% as a long run relative frequency. Donny says: "it will rain in 30% of the city tomorrow". You ask him to elaborate; he says: "Well, there are many different locations in the city. In some of the locations it will rain, in some it won't. It will rain in 30% of the locations, and not in the other 70%. That is, rain will cover 30% of the area of the city, and the other 70% won't have rain." Do you agree? If not, how would you interpret the 30% as a long run relative frequency?
3. You ask Donny Don't to interpret the values 30% and 60% as relative degrees of likelihood. Donny says: "Well, 30% is not that big, so it's not going to rain that hard tomorrow. Also, 60% is twice as big as 30%, so it's going to rain twice as hard two days from now as it will tomorrow". Do you agree? If not, how would you interpret the 30% and 60% as relative degrees of likelihood?
4. Donny says: "Can't we just look at the data from all the days with weather conditions similar to the ones forecast for tomorrow, and see how often it rained on those days to find the probability of rain tomorrow? No subjectivity about that!" How would you respond?

- A **probabilistic forecast** combines observed data and statistical or mathematical models to make predictions.
- Rather than providing a single prediction such as “it will rain tomorrow”, probabilistic forecasts provide a range of scenarios and their relative likelihoods.
- Such forecasts are subjective in nature, relying upon the data used and assumptions of the model.
- Changing the data or assumptions can result in different forecasts and probabilities.

2 Probability Models

- Due to the wide variety of types of random phenomena, an **outcome** can be virtually anything. In particular, an outcome does *not* have to be a number.
- The **sample space** is the set of all possible outcomes of the random phenomenon..
- An event is something that could happen. That is, an *event* is a collection of outcomes that satisfy some criteria. If the sample space outcomes are represented by rows in a spreadsheet, then an event is a subset of rows that satisfies some criteria
- Events are typically denoted with capital letters near the start of the alphabet, with or without subscripts (e.g. A , B , C , A_1 , A_2). Events can be composed from others using [basic set operations](#) like unions ($A \cup B$), intersections ($A \cap B$), and complements (A^c).
 - Read A^c as “not A ”.
 - Read $A \cap B$ as “ A and B ”
 - Read $A \cup B$ as “ A or B ”. Note that unions ($\cup$, “or”) are always inclusive. $A \cup B$ occurs if A occurs but B does not, B occurs but A does not, or both A and B occur.
- A collection of events $A_1, A_2, \dots$ are **disjoint** (a.k.a. mutually exclusive) if $A_i \cap A_j = \emptyset$ for all $i \neq j$. That is, multiple events are disjoint if none of the events have any outcomes in common.
- A **probability measure**, typically denoted P , assigns probabilities to *events* to quantify their relative likelihoods according to the assumptions of the model of the random phenomenon.
- The probability of event A , computed according to probability measure $P(A)$, is denoted $P(A)$.
- A valid probability measure P must satisfy the following three logical consistency “axioms”.
 - For any event A , $0 \leq P(A) \leq 1$.
 - If Ω represents the sample space then $P(\Omega) = 1$.
 - (*Countable additivity.*) If $A_1, A_2, A_3, \dots$ are disjoint then

$$P(A_1 \cup A_2 \cup A_3 \cup \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$$

- Additional properties of a probability measure follow from the axioms
 - *Complement rule.* For any event A , $P(A^c) = 1 - P(A)$.
 - *Subset rule.* If $A \subseteq B$ then $P(A) \leq P(B)$.
 - *Addition rule for two events.* If A and B are any two events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- *Law of total probability.* If $C_1, C_2, C_3 \dots$ are disjoint events with $C_1 \cup C_2 \cup C_3 \cup \dots = \Omega$, then

$$P(A) = P(A \cap C_1) + P(A \cap C_2) + P(A \cap C_3) + \dots$$

- The axioms of a probability measure are minimal logical consistent requirements that ensure that probabilities of different events fit together in a valid, coherent way.
- A single probability measure corresponds to a particular set of assumptions about the random phenomenon.

Example 2.1. Roll a four-sided die twice, and record the result of each roll in sequence. For example, a 3 on the first roll and a 1 on the second is not the same outcome as a 1 on the first roll and a 3 on the second.

1. Identify the sample space.
2. We might be interested in the sum of the two rolls. Explain why it is still advantageous to define the sample space as in the previous part, rather than as just $2, \dots, 8$.

Example 2.2. Regina and Cady are meeting for lunch. They will each definitely arrive between noon and 1, but their exact arrival times are uncertain. Rather than dealing with clock time, it is helpful to represent noon as time 0 and measure time as minutes after noon, including fractions of a minute, so that arrival times take values in the continuous interval $[0, 60]$.

Draw a picture to represent the sample space.

Example 2.3. Roll a four-sided die twice, and record the result of each roll in sequence. Using the sample space from Example 2.1, identify the following events.

1. A , the event that the sum of the two dice is at most 4.
2. B , the event that the larger of the two rolls (or the common roll if a tie) is 3.
3. B^c (identify and interpret).
4. $A \cap B$ (identify and interpret).
5. C , the event that the first roll is a 3.

Example 2.4. Regina and Cady are meeting for lunch. They will each definitely arrive between noon and 1, but their exact arrival times are uncertain. Rather than dealing with clock time, it is helpful to represent noon as time 0 and measure time as minutes after noon, including fractions of a minute, so that arrival times take values in the continuous interval $[0, 60]$.

Using the sample space from Example 2.2, draw a picture to represent each of the following events.

1. A , the event that Regina arrives after Cady.

2. B , the event that either Regina or Cady arrives before 12:30.

3. C , the event that they arrive within 15 minutes of each other.

4. D , the event that Regina arrives before 12:24.

- For a sample space Ω with finitely many possible outcomes, assuming **equally likely outcomes** corresponds to a probability measure P which satisfies

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } \Omega} \quad \text{when outcomes are equally likely}$$

- Note that equally likely outcomes is an *assumption*, and by no means a requirement. In most situations equally likely outcomes is *not* a reasonable assumption.

Example 2.5. Roll a fair four-sided die twice, and record the result of each roll in sequence. Let P represent the probability measure which assumes equally likely outcomes.

1. How many possible outcomes are there? Why is it reasonable to assume they are equally likely?

2. Compute $P(A)$, where A is the event that the sum of the two dice is at most 4.

3. Compute $P(B)$, where B the event that the larger of the two rolls (or the common roll if a tie) is 3.

4. Compute and interpret $P(A \cap B)$. (Is it equal to $P(A)P(B)$?)

5. Compute $P(C)$, where C the event that the first roll is a 3.

- The continuous analog of equally likely outcomes is a **uniform probability measure**. When the sample space is uncountable, size is measured continuously (length, area, volume) rather than discretely (counting).

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{size of } A}{\text{size of } \Omega} \quad \text{if } P \text{ is a uniform probability measure}$$

- Note that a uniform probability measure is an *assumption*, and by no means a requirement. In most situations a uniform probability measure is *not* a reasonable assumption.

Example 2.6. Regina and Cady are meeting for lunch. Suppose they each arrive uniformly at random at a time between noon and 1:00, independently of each other. Record their arrival times as minutes after noon, so noon corresponds to 0 and 1:00 to 60.

Recall the sample space from Example 2.2 and assume a uniform probability measure P .

1. Find the probability that Regina arrives after Cady.

2. Find the probability that either Regina or Cady arrives before 12:30.
3. Find the probability that Cady arrives first and Regina arrives at most 15 minutes after Cady.
4. Find the probability that Regina arrives before 12:24.

2.1 Exercises

Exercise 2.1. The probability that a randomly selected U.S. household has a pet dog is 0.47. The probability that a randomly selected U.S. household has a pet cat is 0.25. (These values are based on the 2018 [General Social Survey \(GSS\)](#).)

1. Represent the information provided using proper symbols.
2. Donny Don't says: "the probability that a randomly selected U.S. household has a pet dog OR a pet cat is $0.47 + 0.25 = 0.72$." Do you agree? What must be true for Donny to be correct? Explain. (Hint: for the remaining parts it helps to consider two-way tables.)

3. What is the *smallest possible* value of the probability that a randomly selected U.S. household has a pet dog AND a pet cat? Describe the (unrealistic) situation in which this extreme case would occur.

4. What is the *largest possible* value of the probability that a randomly selected U.S. household has a pet dog AND a pet cat? Describe the (unrealistic) situation in which this extreme case would occur. What would be the probability that a randomly selected U.S. household has a pet dog OR a pet cat in this scenario?

5. Donny Don't says: "I remember hearing once that in probability OR means add and AND means multiply. So the probability that a randomly selected U.S. household has a pet dog AND a pet cat is $0.47 \times 0.25 = 0.1175$." Do you agree? Explain.

6. According to the GSS, the probability that a randomly selected U.S. household has a pet dog AND a pet cat is 0.15. Compute the probability that a randomly selected U.S. household has a pet dog OR a pet cat.

7. Compute and interpret $P(C \cap D^c)$.

Exercise 2.2. Each question on a multiple choice test has four options. You know with certainty the correct answers to 70% of the questions. For 20% of the questions, you can eliminate two of the incorrect choices with certainty, but you guess at random among the remaining two options. For the remaining 10% of questions, you have no idea and guess one of the four options at random.

1. Randomly select a question from this test. What is the probability that you answer the question correctly? (Hint: make a two-way table.)
2. For any given question on the exam, your probability of answering it correctly is either 1, 0.5, or 0.25, depending on if you know it, can eliminate two choices, or are just guessing. How does your probability of correctly answering a randomly selected question relate to these three values? Which value — 1, 0.5, or 0.25 — is the overall probability closest to, and why?

3 Conditional Probability

Example 3.1. The probability¹ that a randomly selected American adult (18+) uses Snapchat is 0.24.

1. Suppose the randomly selected adult is under age 30. Do you think the probability that a randomly selected *adult who is under age 30* uses Snapchat is 0.24? What if the adult is over age 30?
2. The probability² that a randomly selected American adult is under age 30 is 0.20. Start to create a two-way table representing 10000 hypothetical U.S. adults. Is the probability that a randomly selected American adult both (1) is under age 30, *and* (2) uses Snapchat equal to 0.20×0.24 ? Explain.
3. Suppose that the probability that a randomly selected American adult both is use age 30 and uses Snapchat is 0.13. Construct an appropriate two-way table.
4. Find the probability that a randomly selected American *adult who is under age 30* uses Snapchat.

¹The values in this problem are based on a [April, 2021 report by the Pew Research Center](#).

²Based on data from the [U.S. Census Bureau](#)

5. How can the probability in the previous part be written in terms of the probabilities provided earlier?

6. Find the probability that a randomly selected American *adult who is over age 30* uses Snapchat.

7. Find the probability that a randomly selected American *adult who uses Snapchat* is under age 30.

- The **conditional probability of event** A given event B , denoted $P(A|B)$, is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- The conditional probability $P(A|B)$ represents how the likelihood or degree of uncertainty of event A should be updated to reflect information that event B has occurred.
- Conditional probabilities are probabilities. There are analagous conditional versions of the properties of probability, e.g. $P(A^c|B) = 1 - P(A|B)$.
- In general, knowing whether or not event B occurs influences the probability of event A . That is,

$$\text{In general, } P(A|B) \neq P(A)$$

- Be careful: order is essential in conditioning. That is,

$$\text{In general, } P(A|B) \neq P(B|A)$$

- Within the context of two events, we have joint, conditional, and marginal probabilities.
 - Joint: unconditional probability involving both events, $P(A \cap B)$.
 - Conditional: conditional probability of one event given the other, $P(A|B)$, $P(B|A)$.
 - Marginal: unconditional probability of a single event $P(A)$, $P(B)$.
- The relationship $P(A|B) = P(A \cap B)/P(B)$ can be stated generically as

$$\text{conditional} = \frac{\text{joint}}{\text{marginal}}$$

- When dealing with probabilities (or proportions or percentages) be sure to ask “probability *of what?*” Thinking in fraction terms, be careful to identify the correct reference group which corresponds to the denominator.

Example 3.2. The following two phrases contain exactly the same words, just in different orders. Which is larger, the numerical value of 1 or 2?

1. The probability that a randomly selected man who is greater than six feet tall plays in the National Basketball Association (NBA).
2. The probability that a randomly selected man who plays in the National Basketball Association (NBA) is greater than six feet tall.

Example 3.3. Suppose

- 20% of adults are under age 30
 - 63% of adults under age 30 use TikTok
 - 31% of adults over age 30 use TikTok
1. Use the information to set up a hypothetical two-way table.

2. Compute the probability that an adult uses TikTok.

3. Compute the probability that an adult who uses TikTok is under age 30.

- Rearranging the definition of conditional probability we get the **Multiplication rule**: the probability that two events both occur is

$$\begin{aligned}P(A \cap B) &= P(A|B)P(B) \\ &= P(B|A)P(A)\end{aligned}$$

- The multiplication rule says that you should think “multiply” when you see “and”. However, be careful about *what* you are multiplying: to find a joint probability you need an unconditional and an appropriate conditional probability.
- You can condition either on A or on B , provided you have the appropriate marginal probability; often, conditioning one way is easier than the other.
- Be careful: the multiplication rule does *not* say that $P(A \cap B)$ is the same as $P(A)P(B)$.
- **Law of total probability** If $C_1, C_2, C_3 \dots$ are disjoint events with $C_1 \cup C_2 \cup C_3 \cup \dots = \Omega$, then

$$\begin{aligned}P(A) &= P(A \cap C_1) + P(A \cap C_2) + P(A \cap C_3) + \dots \\ &= P(A|C_1)P(C_1) + P(A|C_2)P(C_2) + P(A|C_3)P(C_3) + \dots\end{aligned}$$

- The events $C_1, C_2, C_3, \dots$, which represent the “cases”, form a *partition* of the sample space; each outcome $\omega \in \Omega$ lies in exactly one of the C_i .
- The law of total probability says that we can interpret the unconditional probability $P(A)$ as a probability-weighted average of the case-by-case conditional probabilities $P(A|C_i)$ where the weights $P(C_i)$ represent the probability of encountering each case.

3.1 Exercises

Exercise 3.1. Continuing Exercise [2.1](#). The probability that a randomly selected U.S. household has a pet dog is 0.47. The probability that a randomly selected U.S. household has a pet cat is 0.25. the probability that a randomly selected U.S. household has a pet dog and a pet cat is 0.15.

In addition to computing, represent the probabilities using appropriate notation.

1. Compute the probability that a randomly selected U.S. household that has a pet dog also has a pet cat. Is it 0.25?
2. Compute the probability that a randomly selected U.S. household that does not have a pet dog has a pet cat. Is it the same as the previous part? Is it one minus the previous part?
3. Describe in words the probability that results from subtracting the answer to the first part from 1.
4. Compute the probability that a randomly selected U.S. household that has a pet cat also has a pet dog.

Exercise 3.2. Suppose that³

- 67% of Democrats believe in human-driven climate

³Probabilities are estimated based on this [2024 survey](#).

- 46% of Independents believe in human-driven climate
- 34% of Republicans believe in human-driven climate

Also suppose that⁴

- 28% of American adults are Democrats
- 42% of American adults are Independents
- 30% of American adults are Republicans

1. Define the event A to represent “believes in human-driven climate change” and D, I, R to correspond to affiliation in each of the parties. If the probability measure P corresponds to selecting an American adult uniformly at random, write all the percentages above as probabilities using proper notation.
2. Construct an appropriate two-way table of probabilities.
3. Now suppose that the randomly selected American believes in human-driven climate change. How does this information change the probability that the selected American belongs to each political party? Answer by computing appropriate probabilities (and using proper notation).
4. How many times more likely is it for an American *adult* to believe in human-driven climate change and be Independent than to:
 - a. believe in human-driven climate change and be Democrat

⁴Estimate based on [Gallup poll](#)

- b. believe in human-driven climate change and be Republican

- 5. How many times more likely is it for an American adult *who believes in human-driven climate change* to be Independent than to be:
 - a. Democrat

 - b. Republican

- 6. What do you notice about the answers to the two previous parts?

- The process of conditioning can be thought of as “**slicing and renormalizing**”.
 - Extract the “slice” corresponding to the event being conditioned on (and discard the rest). For example, a slice might correspond to a particular row or column of a two-way table.
 - “Renormalize” the values in the slice so that corresponding probabilities add up to 1.
- Slicing determines the *shape*; renormalizing determines the *scale*.
- Slicing determines relative probabilities; renormalizing just makes sure they add up to 1.

Exercise 3.3. Continuing Exercise 3.2. Explain in detail how you could conduct a simulation, using only the information provided in the problem set up, and how you would use the results to approximate the conditional probability of each party given belief in human driven climate change.

- Conditional probabilities can be approximated by filtering (or subsetting) based on the condition
- To approximate $P(A|B)$, simulate the random phenomenon for a set number of repetitions (say 10000), *discard those repetitions on which B does not occur*, and compute the relative frequency of A among the remaining repetitions (on which B does occur).
- Be careful! The margin of error is based on only the number of repetitions used to compute the relative frequency. So if you perform 10000 repetitions but B occurs only on 2000, then the margin of error for estimate $P(A|B)$ is roughly on the order of $1/\sqrt{2000} = 0.022$ (rather than $1/\sqrt{10000} = 0.01$. Especially if $P(B)$ is small, the margin of error could be large resulting in an imprecise estimate of $P(A|B)$. (Advantage: not computationally intensive.)

4 Bayes Rule

- *Bayes' rule* describes how to update uncertainty in light of new information, evidence, or data.

Example 4.1. There exist multiple penguin species throughout Antarctica, including the *Adelie* (A), *Chinstrap* (C), and *Gentoo* (G). Suppose that among Antarctic penguins¹ of these three species, 44.2% of Adelie, 19.8% are Chinstrap, and 36.0% are Gentoo. Suppose that

- 83.4% of Adelie penguins are below average weight,
- 89.7% of Chinstrap penguins are below average weight, and
- 4.9% of Gentoo penguins are below average weight (that is, below 4200g).

1. Use the information to construct an appropriate two-way table of probabilities.
2. Compute the probability that a randomly selected penguin is below average weight. How is this related to the values 0.834, 0.897, 0.049?
3. If the selected penguin is below average weight, what is the probability that it is species A? Species C? Species G?

¹Throughout “penguin” refers to an Antarctic penguin of one of these three species. This example is based on the famous [Palmer penguins data set](#).

- **Bayes' rule for events** specifies how a prior probability $P(H)$ of event H is updated in response to the evidence E to obtain the posterior probability $P(H|E)$.

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)}$$

- Event H represents a particular hypothesis (or model or case)
- Event E represents observed evidence (or data or information)
- $P(H)$ is the unconditional or **prior probability** of H (prior to observing evidence E)
- $P(H|E)$ is the conditional or **posterior probability** of H after observing evidence E .
- $P(E|H)$ is the **likelihood** of evidence E given hypothesis (or model or case) H

Example 4.2. Continuing Example 4.1. Randomly select a penguin and suppose it is below average weight. We are interested in classifying the species of the penguin.

1. Frame this problem in the context of Bayes rule: Identify the hypotheses, prior probabilities, evidence, likelihood, and posterior probabilities.
2. Prior to observing the penguin's weight, how many times more likely is it to be Adelie than to be Gentoo?
3. How many times more likely is it for a penguin to be below average weight given that it is Adelie than given that it is Gentoo?
4. Compute the posterior probability of each species given that the penguin is below average weight. Given that the penguin is below average weight, how many times more likely is it to be Adelie than to be Gentoo?

5. How are the ratios in the three previous parts related?

6. Apply the same logic to compare species A and C.

- Bayes rule is often used when there are multiple hypotheses or cases. Suppose $H_1, \dots, H_k$ is a series of distinct hypotheses which together account for all possibilities, and E is any event (evidence).
- Combining Bayes' rule with the law of total probability,

$$\begin{aligned} P(H_j|E) &= \frac{P(E|H_j)P(H_j)}{P(E)} \\ &= \frac{P(E|H_j)P(H_j)}{\sum_{i=1}^k P(E|H_i)P(H_i)} \end{aligned}$$

$$P(H_j|E) \propto P(E|H_j)P(H_j)$$

- The symbol $\propto$ is read “is proportional to”. The relative *ratios* of the posterior probabilities of different hypotheses are determined by the product of the prior probabilities and the likelihoods, $P(E|H_j)P(H_j)$. The marginal probability of the evidence, $P(E)$, in the denominator simply normalizes the numerators to ensure that the updated probabilities sum to 1 over all the distinct hypotheses.
- **In short, Bayes' rule says**

$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

Example 4.3. Continuing Example 4.2. Randomly select a penguin. We are interested in classifying the species of the penguin depending on whether or not it is below average weight.

1. Before you know the weight of the penguin (or any other information) what specify would you predict it to be? Why? What is the probability that your classification is correct?

2. If the penguin is below average weight, what species would you classify it as? Why?

 3. Now suppose that the penguin is not below average weight. Use a Bayes table to compute the posterior probability of each species given that the penguin is not below average weight. If the penguin is not below average weight, what species would you classify it as? Why?

 4. Suppose that you classify any randomly selected penguin based on whether or not it is below average weight as in the two previous parts. What is the posterior probability that you classify the penguin correctly? How does this compare to your probability of being correct before you observed the penguin's weight?
-
- Bayesian analysis is often an iterative process.
 - Posterior probabilities are updated after observing some information or data. These probabilities can then be used as prior probabilities before observing new data.
 - Posterior probabilities can be sequentially updated as new data becomes available, with the posterior probabilities after the previous stage serving as the prior probabilities for the next stage.
 - The final posterior probabilities only depend upon the cumulative data. It doesn't matter if we sequentially update the posterior after each new piece of data or only once after all the data is available; the final posterior probabilities will be the same either way. Also, the final posterior probabilities are not impacted by the order in which the data are observed.

4.1 Exercises

Exercise 4.1. A recent survey of American adults asked: “Based on what you have heard or read, which of the following two statements best describes the scientific method?”

- 70% selected “The scientific method produces findings meant to be continually tested and updated over time”. (We’ll call this the “iterative” opinion.)
- 14% selected “The scientific method identifies unchanging core principles and truths”. (We’ll call this the “unchanging” opinion).
- 16% were not sure which of the two statements was best.

How does the response to this question change based on education level? Suppose education level is classified as: high school or less (HS), some college but no Bachelor’s degree (college), Bachelor’s degree (Bachelor’s), or postgraduate degree (postgraduate). The education breakdown is

- Among those who agree with “iterative”: 31.3% HS, 27.6% college, 22.9% Bachelor’s, and 18.2% postgraduate.
 - Among those who agree with “unchanging”: 38.6% HS, 31.4% college, 19.7% Bachelor’s, and 10.3% postgraduate.
 - Among those “not sure”: 57.3% HS, 27.2% college, 9.7% Bachelor’s, and 5.8% postgraduate
1. Consider the conditional probability that a randomly selected American adult agrees that the scientific method is “iterative” given that they have a postgraduate degree. Identify the prior probability, hypothesis, evidence, likelihood, and posterior probability, and use Bayes’ rule to compute the posterior probability.
 2. Find the conditional probability that a randomly selected American adult with a postgraduate degree agrees that the scientific method is “unchanging”.
 3. Find the conditional probability that a randomly selected American adult with a postgraduate degree is not sure about which statement is best.

4. How many times more likely is it for an *American adult* to agree with the “iterative” statement than to agree with the “unchanging” statement?

5. How many times more likely is it for an American adult to have a postgraduate degree when the adult agrees with the iterative statement than when the adult agree with the unchanging statement?

6. How many times more likely is it for an *American adult with a postgraduate degree* to agree with the “iterative” statement than to agree with the “unchanging” statement?

7. How are the values in the three previous parts related?

Exercise 4.2. Now suppose we want to compute the posterior probabilities for an American adult’s perception of the scientific method given that the randomly selected American adult has some college but no Bachelor’s degree (“college”).

1. Before computing, make an educated guess for the posterior probabilities. In particular, will the changes from prior to posterior be more or less extreme given the American has some college but no Bachelor’s degree than when given the American has a postgraduate degree? Why?

2. Construct a Bayes table and compute the posterior probabilities. Compare to the posterior probabilities given postgraduate degree from the previous examples.

Exercise 4.3. Within both the colleges of Agriculture and Architecture at Cal Poly, about 49% of admitted students are female, about 84% of admitted students went to high school in CA, and the median GPA of admitted students is about 4.1.

An orientation group of 100 newly admitted Cal Poly students includes 75 students in Agriculture and 25 students in Architecture. A student is randomly selected from this group. The selected student is Maddie, who is female, went to high school in CA, and had a high school GPA of 4.1.

Donny Don't says, "The information about Maddie applies equally well to Agriculture or Architecture and doesn't help us decide which college she's in, so it's just 50/50. Given the information about Maddie, the conditional probability that she is in Agriculture is 0.5." Do you agree? If not, what is the conditional probability that Maddie is in the college of Agriculture given the information about her?

5 Independence of Events

Example 5.1. Do Americans think it is important for federal judges to be impartial in how they decide court cases? Consider the following data from the [Pew Research Center](#). (Independents are classified as either lean Democrat or lean Republican.)

	Democrat/lean Dem (D)	Republican/lean Rep (D^c)	Total
Important (I)	1674	1531	3205
Not important (I^c)	146	133	279
Total	1820	1664	3484

Suppose a person is randomly selected from this sample. Consider the events

$$I = \{\text{person thinks it is important for judges to be impartial}\}$$

$$D = \{\text{person is a Democrat/leans Dem}\}$$

1. Compute and interpret $P(I)$.
2. Compute and interpret $P(I|D)$.
3. Compute and interpret $P(I|D^c)$.

4. What do you notice about $P(I)$, $P(I|D)$, and $P(I|D^c)$?
5. Compute and interpret $P(D)$.
6. Compute and interpret $P(D|I)$.
7. Compute and interpret $P(D|I^c)$.
8. What do you notice about $P(D)$, $P(D|I)$, and $P(D|I^c)$?
9. Compute and interpret $P(D \cap I)$.
10. What is the relationship between $P(D \cap I)$ and $P(D)$ and $P(I)$?
11. When randomly selecting a person from this particular group, would you say that events

D and I are independent? Why?

- Events A and B are **independent** if the knowing whether or not one occurs does not change the probability of the other.
- For events A and B (with $0 < P(A) < 1$ and $0 < P(B) < 1$) the following are equivalent. That is, if one is true then they all are true; if one is false, then they all are false.

A and B are independent

$$P(A \cap B) = P(A)P(B)$$

$$P(A^c \cap B) = P(A^c)P(B)$$

$$P(A \cap B^c) = P(A)P(B^c)$$

$$P(A^c \cap B^c) = P(A^c)P(B^c)$$

$$P(A|B) = P(A)$$

$$P(A|B) = P(A|B^c)$$

$$P(B|A) = P(B)$$

$$P(B|A) = P(B|A^c)$$

Example 5.2. This is a simplified example of *batch testing*¹ for a disease. In batch testing, specimens (like blood or saliva samples) from a group of people are pooled together into one batch, which then undergoes one test. If none of the people has the disease, then the batch test result will be negative, and no further tests are required. But if at least one person in the batch has the disease, then the batch test result will be positive, and then each person in the batch must be tested individually. (We're assuming no false positives and no false negatives for this example.)

Suppose that 8 people are to be tested, first in a single batch. Assume that each person has probability $p = 0.1$ of having the disease, independently from person to person. (The assumption of independence might be unreasonable if the people are in close contact with one another.)

1. Identify the possible values for the total number of tests conducted.

¹Thanks to Allan Rossman for this example. His [blog](#) has more investigations into batch testing, including what happens as n and p change. [Cal Poly](#) implemented versions of batch testing for COVID through [saliva samples](#) and [wastewater testing](#).

2. Compute the probability of each possible value.

- Events $A_1, A_2, A_3, \dots$ are **independent** if:
 - any pair of events $A_i, A_j, (i \neq j)$ satisfies $P(A_i \cap A_j) = P(A_i)P(A_j)$,
 - and any triple of events A_i, A_j, A_k (distinct i, j, k) satisfies $P(A_i \cap A_j \cap A_k) = P(A_i)P(A_j)P(A_k)$,
 - and any quadruple of events satisfies $P(A_i \cap A_j \cap A_k \cap A_m) = P(A_i)P(A_j)P(A_k)P(A_m)$,
 - and so on.
- Intuitively, a collection of events is independent if knowing whether or not any combination of the events in the collection occur does not change the probability of any other event in the collection.
- When events are independent, the multiplication rule simplifies greatly.

$$P(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n) = P(A_1)P(A_2)P(A_3) \dots P(A_n) \quad \text{if } A_1, A_2, A_3, \dots, A_n \text{ are independent}$$

- When a problem involves independence, you will want to take advantage of it. Work with “and” events whenever possible in order to use the multiplication rule. For example, for problems involving “at least one” (an “or” event) take the complement to obtain “none” (an “and” event).

Example 5.3. Continuing Example 5.2. Now suppose that 8 people to be tested are split into two groups of 4 people each. Within each group of 4 people, specimens are combined into a batch to be tested. If anyone in the batch has the disease, then the batch test will be positive, and those 4 people will need to be tested individually. Assume that each person has probability $p = 0.1$ of having the disease, independently from person to person.

1. Identify the possible values for the total number of tests conducted.

2. Compute the probability of each possible value.

5.1 Exercises

Exercise 5.1. A certain system consists of four identical components. Suppose that the probability that any particular component fails is 0.1, and failures of the components occur independently of each other. Find the probability that the system fails if:

1. The components are connected in *parallel*: the system fails only if *all* of the components fail.
2. The components are connected in *series*: the system fails whenever *at least one* of the components fails.
3. Donny Don't says the answer to the previous part is $0.1 + 0.1 + 0.1 + 0.1 = 0.4$. Explain the error in Donny's reasoning.

Exercise 5.2. Roll two fair four-sided dice, one green and one gold. There are 16 total possible outcomes (roll on green, roll on gold), all equally likely. Consider the event $E = \{\text{the green die lands on 1}\}$. Answer the following questions by computing and comparing appropriate probabilities.

1. Consider $A = \{\text{the gold die lands on 4}\}$. Are A and E independent?

2. Consider $B = \{\text{the sum of the dice is 3}\}$. Are B and E independent?

3. Consider $C = \{\text{the sum of the dice is 5}\}$. Are C and E independent?

- Independence concerns whether or not the occurrence of one event affects the *probability* of the other.
- Given two events it is not always obvious whether or not they are independent.
- Independence depends on the underlying probability measure. Events that are independent under one probability measure might not be independent under another.
- Independence is often assumed. Whether or not independence is a valid assumption depends on the underlying random phenomenon.

6 Discrete Random Variables

- Roughly, a *random variable* assigns a number measuring some quantity of interest to each outcome of a random phenomenon.
- Mathematically, a **random variable (RV)** X is a *function* that takes an outcome in the sample space as input and returns a real number as output
- The random variable itself is typically denoted with a capital letter (X); possible values of that random variable are denoted with lower case letters (x).
 - Think of the capital letter X as a label standing in for a formula like “the number of heads in 4 flips of a coin” and
 - x as a dummy variable standing in for a particular value like 3.
- **Discrete random variables** take at most countably many possible values (e.g., 0, 1, 2, ...). They are often counting variables (e.g., the number of Heads in 10 coin flips).
- **Continuous random variables** can take any real value in some interval (e.g., $[0, 1]$, $[0, \infty)$, $(-\infty, \infty)$). That is, continuous random variables can take uncountably many different values. Continuous random variables are often measurement variables (e.g., height, weight, income).
- A function of a random variable is also a random variable: if X is a RV then so is $g(X)$
- Sums and products, etc., of random variables *defined on the same sample space* are random variables. If X and Y are RVs defined on the same sample space then so are $X + Y$, $X - Y$, XY
- If the sample space outcomes are represented by rows in a spreadsheet, then random variables correspond to columns.
- Expressions like $X = x$ or $\{X = x\}$ represent *events*: for which outcomes is the value of the random variable X equal to the value x . (Remember, if the sample space outcomes are represented by rows in a spreadsheet, then an event corresponds to a subset of rows (outcomes) that satisfies some criteria.)
- The **(probability) distribution** of a collection of random variables identifies the possible values that the random variables can take and their relative likelihoods.
- We will see many ways of describing a distribution, depending on how many random variables are involved and their types (discrete or continuous).

Table 6.1: Two rolls of a fair four-sided die

First roll	Second roll	X	Y
1	1		
1	2		
1	3		
1	4		
2	1		
2	2		
2	3		
2	4		
3	1		
3	2		
3	3		
3	4		
4	1		
4	2		
4	3		
4	4		

Example 6.1. Roll a four-sided die twice, and record the result of each roll in sequence. Let X be the sum of the two dice, and let Y be the larger of the two rolls (or the common value if both rolls are the same). Table 6.1 represents the possible outcomes. Assume the die is fair and the rolls are independent.

1. Fill in the values of X and Y for each outcome in the table.
2. Identify the event $\{X = 4\}$ and compute its probability. Then interpret the probability both as a long relative frequency and as a relative likelihood.
3. Construct a table and plot of $P(X = x)$ for each possible value x of X .

4. Construct a table and plot of $P(Y = y)$ for each possible value y of Y .

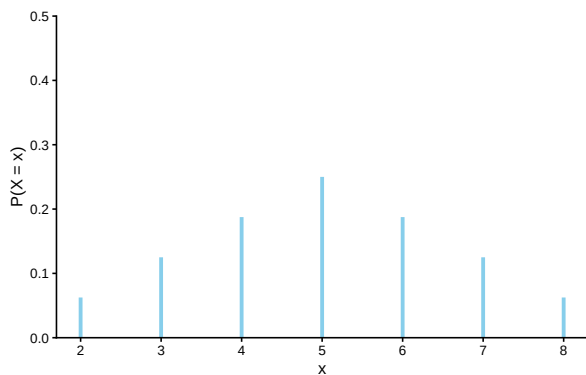


Figure 6.1: The marginal distribution of X , the sum of two rolls of a fair four-sided die.

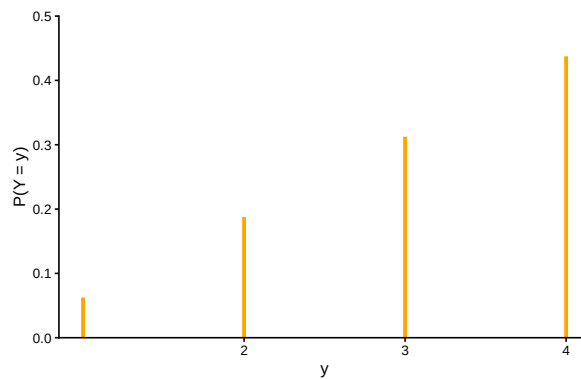


Figure 6.2: The marginal distribution of Y , the larger of two rolls of a fair four-sided die.

Example 6.2. Recall the batch testing scenarios from Example 5.2 and Example 5.3. Let X be the total number of tests required when we test all 8 people in a single batch first, and let Y be the total number of tests when we split into two batches of 4 people first.

1. Identify the distribution of X .

2. Identify the distribution of Y .

Example 6.3. The “matching problem” involves n distinct objects labeled $1, \dots, n$ which are placed in n distinct boxes labeled $1, \dots, n$, with exactly one object placed in each box. Suppose the objects are placed in the boxes uniformly at random, so that any possible arrangement is equally likely. Let X be the number of matches, that is, the number of objects for which their label matches the label of the box in which they are placed.

For example, suppose n people have a secret Santa gift exchange. Each person puts their name in a hat, the names are well shuffled, and everyone draws a name from the hat (to then purchase a gift for the selected person). Let X represent the number of people who draw their own name.

Consider the matching problem with $n = 3$. Specify the distribution of X , and interpret it.

6.1 Exercises

Exercise 6.1. Five individuals are interviewing for a single job opening, but only two are qualified for the position. Suppose the individuals are interviewed in a random order, and each individual is only interviewed once, until one of the qualified candidates is interviewed, and then no more candidates are interviewed. Let X count the number of interviews conducted.

Make a table to represent the distribution of X .

Exercise 6.2. Continuing Example 6.1.

1. Describe how you could simulate a single value of Y .
 2. Construct a spinner (like from a kids game) to represent the distribution of Y .
 3. Describe another way to simulate a single value of Y .
 4. Describe how you could use simulation to approximate the distribution of Y .
- Don't confuse a random variable with its distribution!
 - A random variable measures a numerical quantity which depends on the outcome of a random phenomenon
 - The distribution of a random variable specifies the long run pattern of variation of values of the random variable over many repetitions of the underlying random phenomenon.
 - Any marginal distribution can be represented by a single spinner (like from a kids game).
 - In principle, there are always two ways of simulating a value x of a random variable X .
 1. *Simulate from the probability space.* Simulate an outcome ω from the underlying probability space and set $x = X(\omega)$.
 2. *Simulate from the distribution.* Construct a spinner corresponding to the distribution of X and spin it once to generate x .
 - The second method requires that the distribution of X is known. However, as we will see in many examples, *it is common to specify the distribution of a random variable directly without defining the underlying probability space.*

Homework 1

1. In each of the following, write a clearly worded sentence interpreting the numerical value of the probability as a long run relative frequency in context. (Just take the numerical values as given for now. We'll see how to compute probabilities like these later.)
 - a. The probability of rolling doubles when you roll two fair six-sided dice is $1/6$.
 - b. The probability of rolling doubles on three consecutive rolls of two fair six-sided dice is 0.00463.
 - c. The probability that the sum of 100 rolls of a fair six-sided die is less than 370 is 12%.
 - d. Roll a fair six-sided die until you roll a 6 three times and then stop. The probability that you roll the die at least 10 times is 0.822.
2. Suppose that at some point in 2023 your subjective probabilities for who would win the 2024 U.S. Presidential Election satisfied the following. (Imagine a time before Trump had been declared the Republican nominee, and while Biden was the presumptive Democratic nominee there was still speculation that he should step aside.)
 - The Democratic nominee and the Republican nominee are equally likely to be the winner of the Presidential Election
 - Joe Biden is 5 times more likely to win than Kamala Harris, and no other Democratic candidate has a chance of winning
 - Donald Trump is twice as likely to win as any other Republican candidate.

Create a table of your subjective probabilities.

1. Suppose that you have applied to two graduate schools, A and B. Your subjective probability of being accepted is 0.6 for school A and 0.7 for school B. Hint: the fact that these are subjective probabilities does not change the way you solve the problem. Remember, the mathematics of probability is the same regardless of whether the probabilities represent long run relative frequencies or subjective degrees of relative likelihood. Just take the 0.6 and 0.7 values as given and use them to solve the following parts.

Let A represent the event that you are accepted at school A, B the event that you are accepted at school B, and P the probability measure that represents your subjective

probabilities. In addition to computing, represent each of the probabilities below with appropriate notation.

- a. Interpret the subjective probabilities: How many times more likely than not are you to be accepted at school A? How many times more likely are you to be accepted at school B than at school A?
 - b. What is the largest possible probability of being accepted by both schools? Under what scenario (however unrealistic) would this be true? Explain.
 - c. What is the smallest possible probability of being accepted by both schools? Under what scenario (however unrealistic) would this be true? Explain.
 - d. Explain why your subjective probability of being accepted by both schools is not necessarily 0.42.
 - e. **For the remaining parts, suppose your subjective probability of being accepted at both schools is 0.55.** If you are accepted at school A, what is your probability of also being accepted at school B? For this and the remaining parts in addition to computing the probability represent it with proper notation.
 - f. If you are accepted at school A, what is your probability of not being accepted at school B?
 - g. If you are not accepted at school A, what is your probability of being accepted at school B?
 - h. If you are accepted at school B, what is your probability of also being accepted at school A?
 - i. If you are not accepted at school B, what is your probability of being accepted at school A?
2. For each of the following, create your own “which of (a) or (b) is larger?” example. You can use the ones from the handout as examples, but you should choose your own contexts. Identify the correct answer and write a sentence or two explaining the answer to a student who is confused. You can use illustrations in your explanations, but don’t do any calculations.
- a. Reverse the direction of conditioning (like the 6 foot tall and NBA example)
 - b. Compare conditional and unconditional probabilities (like man, versus man who is greater than 6 feet tall)
 - c. Two “particulars” that people might think are not equally likely but are (like HHHHH versus HTHTT).
 - d. Comparing “the particular” and “the general” (HTHTT versus 2 H in 5 flips).
 - e. Comparing the probability of happening once versus at least once (like you winning the lottery versus someone winning the lottery)
3. The “matching problem” involves n distincts objects labeled $1, \dots, n$ which are placed in n distinct boxes labeled $1, \dots, n$, with exactly one object placed in each box. Suppose the objects are placed in the boxes uniformly at random, so that any possible arrangement is equally likely. We’re interested in the number of matches (the number of objects for

which their label matches the label of the box in which they are placed.) Let C be the event that at least one object is placed in the correct spot (at least one match).

For example, suppose n people have a secret Santa gift exchange. Each person puts their name in a hat, the names are well shuffled, and everyone draws a name from the hat (to then purchase a gift for the selected person). Then C represents the event that at least one person draw their own name.

- a. Describe in full detail how, in principle, you would physical objects (like cards) to simulate a single repetition of the placement of objects in boxes, and the number of matches.
 - b. Describe in full detail how, in principle, you could conduct a simulation and use the results to approximate $P(C)$.
 - c. [This applet conducts a simulation of the matching problem](#) (in the context of babies in a hospital being mixed up and returned to parents uniformly at random.) In the applet, “Number of babies” represents n and “Number of trials” represents the number of simulated repetitions. Starting with $n = 4$, try different values for the number of repetitions and hit the “Randomize” button to run the simulation. It helps to start with just a few repetitions so that you can see how the simulation is working, before jumping to thousands of repetitions. Use the simulation results to approximate $P(C)$ when $n = 4$.
 - d. How do you expect $P(C)$ to depend on n ? As n increases, do you expect $P(C)$ to increase, decrease, or stay about the same? Just think about it before proceeding; what does your intuition say?
 - e. Now try a few different values of n (say $n = 5$, $n = 10$, and $n = 20$, but you’re free to choose other values). For each value of n run the simulation and use the results to approximate $P(C)$. Record your results.
 - f. What do the simulation results suggest about the relationship between $P(C)$ and n ? (Remember that there is simulation margin of error, based on the number of repetitions. Two numerical approximations that are within the margin of error of each other are essentially the same approximation.)
4. Katniss throws a dart at a circular dartboard with radius 1 foot. Suppose that Katniss’s dart lands at a uniformly random location on the dartboard (and she never misses the dartboard).
 - a. Compute the probability that Katniss’s dart lands within 1 inch of the center of the dartboard.
 - b. Compute the probability that Katniss’s dart lands more than 1 inch but less than 2 inches away from the center of the dartboard.
 5. Imagine a light that flashes every few seconds¹. The light randomly flashes green with probability 0.75 and red with probability 0.25, independently from flash to flash.

¹Thanks to Allan Rossman for this example.

- a. Write down a sequence of G's (for green) and R's (for red) to predict the colors for the next 40 flashes of this light. Before you read on, please take a minute to think about how you would generate such a sequence yourself.
- b. Most people produce a sequence that has 30 G's and 10 R's, or close to those proportions, because they are trying to generate a sequence for which each outcome has a 75% chance for G and a 25% chance for R. That is, they use a strategy in which they predict G with probability 0.75, and R with probability 0.25. How well does this strategy do? Compute the probability of correctly predicting any single item in the sequence using this strategy.
- c. Describe a better strategy. (Hint: can you find a strategy for which the probability of correctly predicting any single flash is 0.75?)

Homework 2

Problem 1

A standard deck of playing cards has 52 cards, 13 cards (2 through 10, jack, king, queen, ace) in each of 4 suits (hearts, diamonds, clubs, spades). Shuffle a deck and deals cards one at a time without replacement.

1. What is the probability that the first card dealt is a heart?
2. What is the probability that the second card dealt is a heart? Just make a guess now before proceeding, but you'll revisit this a few parts down.
3. What is the probability that the second card dealt is a heart if the first card dealt is a heart?
4. What is the probability that the second card dealt is a heart if the first card dealt is not a heart?
5. Revisit part 2. What is the probability that the second card dealt is a heart? Hint: your answer should be a single number, but show all your work.
6. Find the probability that the first two cards dealt are hearts.
7. Find the probability that the first two cards dealt are hearts and the third card dealt is a diamond.

Problem 2

The ELISA test for HIV was widely used in the mid-1990s for screening blood donations. As with most medical diagnostic tests, the ELISA test is not perfect. If a person actually carries the HIV virus, experts estimate that this test gives a positive result 97.7% of the time. (This number is called the *sensitivity* of the test.) If a person does not carry the HIV virus, ELISA gives a negative (correct) result 92.6% of the time (the *specificity* of the test). Estimates at the time were that 0.5% of the American public carried the HIV virus.

Suppose that a randomly selected American tests positive; we are interested in the conditional probability that the person actually carries the virus.

1. Before proceeding, make a guess for the probability in question. Which range do you think it's in: 0-20%, 20-40%, 40-60%, 60-80%, or 80-100%?
2. Denote the probabilities provided in the setup using proper notation

3. Construct an appropriate two-way table and use it to compute the probability of interest.
4. Construct a Bayes table and use it to compute the probability of interest.
5. Explain why this probability is small, compared to the sensitivity and specificity.
6. By what factor has the probability of carrying HIV increased, given a positive test result, as compared to before the test?
7. Now suppose that 5% of individuals in a high-risk group carry the HIV virus. Consider a randomly selected person from this group who takes the test. Given that the test is positive, how many times more likely is it for the person not to have HIV than to have it? Answer without first computing a two-way or Bayes table.
8. Using the result from the previous part, compute the conditional probability that a person in this risk group who tests positive has HIV.
9. Is the posterior probability influenced by the prior probability? Discuss.

Problem 3

Consider three tennis players Arya (“A”), Brienne (“B”), and Cersei (“C”). One of these players is better than the other two, who are equally good/bad. When the best player plays either of the others, she has a $2/3$ probability of winning the match. When the other two players play each other, each has a $1/2$ probability of winning the match. But you do not know which player is the best. Based on watching the players warm up, you start with subjective probabilities of 0.5 that A is the best, 0.35 that B is the best, and 0.15 that C is the best. A and B will play the first match.

1. Suppose that A beats B in the first match. Compute your posterior probability that each of A, B, C is best given that A beats B in the first match.
2. Compare the posterior probabilities from the previous part to the prior probabilities. Explain how your probabilities changed, and why that makes sense.
3. **Coding required.** Code and run a simulation to simulate (1) who is the best player according to your prior distribution, (2) who wins the first match given who is the best player; then use the results to approximate the probabilities from the first part.
4. Suppose instead that B beats A in the first match. Compute your posterior probability that each of A, B, C is best given that B beats A in the first match.
5. Compare the posterior probabilities from the previous part to the prior probabilities. Explain how your probabilities changed, and why that makes sense.
6. Now suppose again that A beats B in the first match, and also that A beats C in the second match. Compute your posterior probability that each of A, B, C is best given the results of the first two matches. (Hint: use as the prior your posterior probabilities from the previous part.) Explain how your probabilities changed, and why that makes sense.

Problem 4

In the Powerball lottery, a player picks five different whole numbers between 1 and 69, and another whole number between 1 and 26 that is called the Powerball. In the drawing, the 5 numbers are drawn without replacement from a “hopper” with balls labeled 1 through 69, but the Powerball is drawn from a separate hopper with balls labeled 1 through 26. The player wins the jackpot if both the first 5 numbers match those drawn, in any order, and the Powerball is a match. Under this set up, there are 292,201,338 possible winning numbers.

1. What is the probability the next winning number is 6-7-16-23-26, plus the Powerball number, 4.
2. What is the probability the next winning number is 1-2-3-4-5, plus the Powerball number, 6.
3. The Powerball drawing happens twice a week. Suppose you play the same Powerball number, twice a week, every week for over 50 years. Let's say you purchase a ticket for 6000 drawings in total. What is the probability that you win at least once?
4. Instead of playing for 50 years, you decide only to play one lottery, but you buy 6000 tickets, each with a different Powerball number. What is the probability that at least one of your tickets wins? How does this compare to the previous part? Why?
5. Each ticket costs 2 dollars, but the jackpot changes from drawing to drawing. Suppose you buy 6000 tickets for a single drawing. How large does the jackpot need to be for your “expected” profit to be positive? To be \$100,000? (We're ignoring inflation, taxes, transaction costs, and any changes in the rules.)

Problem 5

Consider a “best-of-5” series of games between two teams: games are played until one of the teams has won 3 games (requiring at most 5 games total). Suppose one team, team A, is better than the other, having a 0.55 probability of winning any particular game. Assume the results of the games are independent (and ignore advantage, etc). Let X represent the number of games played in the series. (Hint: It's helpful to first construct a two-way table of probabilities with the number of games played and which team wins, and then use it to answer the following questions. It will also help to list some outcomes, like AABA (team A wins game 1, 2, and 4, and B wins game 3).)

1. Compute the probability that team A wins the series in 3 games.
2. Compute the probability that the series ends in 3 games.
3. Compute the probability that team A wins the series.
4. Are the events “team A wins the series” and “the series ends in 3 games” independent? Explain by comparing relevant probabilities.
5. Let X represent the number of games played in the series. Find the distribution of X .

Summary of Some Common Distributions

Name and parameters	Possible values	pmf/pdf	cdf	Mean	SD
Discrete					
Poisson(μ)	$x = 0, 1, 2, \dots$	$e^{-\mu} \frac{\mu^x}{x!}$	No closed form	μ	$\sqrt{\mu}$
Binomial(n, p)	$x = 0, 1, 2, \dots, n$	$\binom{n}{x} p^x (1-p)^{n-x}$	No closed form	np	$\sqrt{np(1-p)}$
Geometric(p)	$x = 1, 2, 3, \dots$	$p(1-p)^{x-1}$	$1 - (1-p)^x$	$\frac{1}{p}$	$\sqrt{\frac{1-p}{p^2}}$
Continuous					
Uniform(a, b)	$a < x < b$	$\frac{1}{b-a}$	$\frac{x-a}{b-a}$	$\frac{a+b}{2}$	$\frac{b-a}{\sqrt{12}}$
Exponential(λ)	$x > 0$	$\lambda e^{-\lambda x}$	$1 - e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda}$
Normal(μ, σ)	$-\infty < x < \infty$	$\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$	No closed form (use tables)	μ	σ

References